

LEO ZHOU

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Education

Rice University

August 2025 – December 2028

Bachelor of Science in Computer Science

Houston, Texas

GPA: 3.92/4.00 **Relevant Coursework:** Data Structures & Algorithms, Linear Algebra, Computer Organization, Discrete Math, Intro to Computer Engineering, Digital Logic Design, Multivariable Calculus

Technical Skills

Languages: Python, C/C++, Swift, JavaScript/TypeScript, SQL (PostgreSQL), HTML/CSS

Web & Backend: React, Node.js, FastAPI, Flask, REST APIs, WebSockets, Redis (Pub/Sub)

Infrastructure: Docker, GitHub Actions (CI/CD), Git, Locust, Sentry

ML / Edge AI: PyTorch, TensorFlow Lite Micro, Quantization (PTQ/QAT), Knowledge Distillation, Pruning

Embedded & Mobile: STM32 (bare-metal), FreeRTOS, ESP32, UART/SPI/I²C, SwiftUI, Apple Vision (OCR)

Experience

CanSemi

June 2026 – Present

Software Engineer Intern

Guangzhou, China

- Built event-driven binary parser extracting parametric test records from 500+ ATE datalogs, used by 6 test engineers
- Designed normalized PostgreSQL schema (lot/wafer/die/result) & batch pipeline loading 40 lots (~12,000 die) in <2 min/lot
- Developed a FastAPI backend with 8 query endpoints serving wafer/lot-level test data, decreasing reporting time by ~75%
- Built an interactive React/Canvas wafer map rendering 300+ die with hover and drill-down, replacing static Excel heatmaps

ABC Reads

May 2026 – Present

Founder & Software Engineer

Remote

- Founded a native SwiftUI iOS app for heritage Chinese speakers, personalized by a per-user Known-Character Model
- Built a real-time camera pipeline (Apple Vision OCR + word segmentation) overlaying pinyin on only unfamiliar characters
- Compiled a 120,000-entry offline SQLite dictionary via a Python pipeline, frequency-ranked against a SUBTLEX-CH corpus
- Designed a local-first data layer over the raw SQLite3 C API powering spaced-repetition review and a WidgetKit widget

Rice Wind Energy

October 2025 – April 2026

Firmware Engineer

Houston, Texas

- Developed ESP32 firmware in C/C++ for a 3-phase wind turbine controller with RPM, power, and pitch-control sensing
- Built a 6-state safety state machine with fault latching/recovery and a perturb-and-observe peak-power tracking algorithm

A-Lab@Rice

November 2025 – February 2026

Undergraduate Researcher

Houston, Texas

- Built COMSOL Multiphysics simulations for a solar thermal desalination project (STREED) to optimize model efficiency
- Executed parameter sweeps and analyzed outputs to improve model performance and support design alterations

Projects

SeatLive — Concurrency-Safe Booking Engine | FastAPI, PostgreSQL, Redis, WebSockets May 2026 – June 2026

- Built a real-time booking engine preventing double-booked seats via atomic conditional UPDATES with Redis TTL holds
- Implemented idempotent confirmation via client-supplied keys, short-circuiting retries instead of double-processing
- Proved correctness with CI concurrency tests firing 50 simultaneous requests at one seat, ensuring exactly one succeeds
- Streamed live seat-map updates via WebSockets + Redis pub/sub; Locust load test served 4,459 requests with zero failures
- Containerized app, Postgres, and Redis with Docker Compose; initialized Sentry to capture unhandled exceptions

BlunderNet — Self-Training Chess Engine | PyTorch, C++, pybind11, GitHub Actions

July 2026 – Present

- Built an AlphaZero-style chess engine (residual CNN with policy/value heads) that retrains itself daily on fresh top-player Lichess games with cursor-based ingest and game-ID dedup, fully automated on scheduled GitHub Actions (5 runs/day)
- Implemented PUCT Monte-Carlo Tree Search in Python and C++ (pybind11, struct-of-arrays tree), batching leaf evaluations via virtual loss for a 1.8× speedup at equal simulations with verified top-move agreement
- Closed the reinforcement loop with daily self-play, training the policy toward MCTS root visit distributions
- Tracked strength with three benchmarks: held-out move prediction, a 1,200-puzzle Lichess tactics suite, and daily Elo estimation vs. Stockfish (~1000 Elo and climbing)

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- Implemented PUCT MCTS in Python and C++ (pybind11), batching leaf evaluations via virtual loss for a 1.8× speedup
- Automated fresh Lichess top-player game ingest with cursor-based pagination and game-ID dedup
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Edge-ML Audio Compression Benchmark | *PyTorch, TensorFlow Lite Micro, C++* **June 2026 – July 2026**

- Built a model-compression pipeline (knowledge distillation, INT8 quantization, magnitude pruning) for a 50-class sound classifier; compressed a 3.9 MB CNN to 71 KB (55× smaller) losing only 6.7 pts accuracy
- Benchmarked PTQ vs. QAT across all variants to produce an accuracy/latency/memory Pareto frontier; implemented distillation-aware QAT recovering full FP32 accuracy at 3.1× compression
- Deployed to an ESP32-S3 with TFLite Micro and a custom C++ mel-spectrogram front end; achieved ~300 ms on-device inference (10× real time) in ~300 KB RAM
- Built a React dashboard charting the Pareto frontier and streaming live on-device inference over WebSockets

Edge-ML Audio Compression Benchmark | *PyTorch, TensorFlow Lite Micro* **June 2026 – July 2026**

- Compressed a 3.9 MB CNN audio classifier (ESC-50) to 71 KB (55×) via distillation → INT8 quantization → pruning
- Benchmarked PTQ vs. QAT on a Pareto frontier; distillation-aware QAT recovered FP32 accuracy at 3.1× compression
- Deployed to an ESP32-S3 with TFLite Micro and a C++ mel-spectrogram front end: ~300 ms inference in ~300 KB RAM

Equity Signal Research Platform | *Python, DuckDB, GitHub Actions* **July 2026 – Present**

- Built a signal research platform with pre-registered hypotheses and kill criteria; idempotent daily OHLCV ingest into an append-only DuckDB store across a ~100-name S&P universe and 26 years of history
- Wrote a vectorized, leakage-tested backtester reporting results gross and net of a 10 bps/side cost model, benchmarked against turnover-matched random-signal null distributions
- Falsified 20-day momentum (gross Sharpe −0.49) and showed the flipped short-term reversal book is exactly cost-breakeven under a Bonferroni-corrected significance bar
- Built a walk-forward harness (rolling 2y train / 6m test) demonstrating out-of-sample parameter tuning underperforms every fixed variant (−0.14 net Sharpe)
- Accumulated a 15,000+ headline archive via scheduled RSS pulls; embedded with a sentence-transformer and tested pre-registered sentiment features under the same cost-aware evaluation

Quant-Signals — Equity Signal Research Platform | *Python, DuckDB, Github Actions* **July 2026 – Present**

- Built a platform with pre-registered hypotheses and kill criteria; daily idempotent OHLCV ingest into append-only DuckDB
- Wrote a vectorized, leakage-tested backtester reporting gross/net of 10 bps/side costs vs. random-signal null baselines
- Falsified 20-day momentum (gross Sharpe −0.49); flipped short-term reversal is cost-breakeven after Bonferroni correction
- Built a walk-forward harness (2y train / 6m test) showing tuned parameters underperform every fixed variant out-of-sample
- Archived 15,000+ RSS headlines, embedded with a sentence-transformer, and tested sentiment under the same cost model

STM32 Bare-Metal Motor Controller | *Embedded C, STM32, FreeRTOS* **March 2026 – July 2026**

- Wrote bare-metal C firmware on a Nucleo-F401RE with HAL-free GPIO, UART, SPI, I2C, and timer drivers from scratch
- Implemented closed-loop PID speed control via PWM and quadrature encoder feedback with <2% steady-state error
- Ported the control task to FreeRTOS; built a UART telemetry pipeline with Python scripts plotting PID step response

Leadership / Extracurricular

Junior National Fencing Team Captain + Coach & Referee **January 2022 – March 2025**

- Captained NZ's junior national team to Oceania titles; competed at Junior Commonwealth Games & World Championships
- Earned 27 national first-place finishes; coached 100+ youth fencers and refereed hundreds of bouts across Oceania